

Table 11: **Detection, repetition within responses, and diversity between responses.** Each method generates two 1,024-token responses for each of 50 prompts using Qwen3-8B-Base at temperature 1 with the full vocabulary. Self-BLEU measures average overlap in wording between the two responses to each prompt; lower values indicate greater between-response diversity. Repeated 4-grams is the percentage of contiguous four-token sequences that have already appeared earlier in the same response, counting every occurrence after the first. Distinct-3 is the number of distinct contiguous three-token sequences divided by the total number of three-token sequences in a response; higher values indicate less repetition. Both metrics include overlapping sequences and are averaged over the 100 responses. TPR is the percentage of watermarked responses detected. Each detector uses only the generated response and targets a false-positive rate of 0.1%. With repeat fallback on, SynthID, TextSeal, and Gumbel-Max use unwatermarked sampling at positions where the watermarking context repeats within a response. With it off, they continue applying the watermarking rule at those positions. Settings for the PRC watermark: online generation, $\eta = 0.05$, check weight $t = 3$, row rate 99/100, and posterior-mean/Hoeffding detection. Small upper and lower scripts give the upper and lower limits of 95% confidence intervals from 2,000 bootstrap resamples of the 50 prompts, keeping each response pair together. The PRC row is bold for reference; other values are bold when the paired 95% interval for their difference from PRC excludes zero, without adjustment for multiple comparisons. At 100% TPR, bootstrap intervals collapse to 100–100; this does not imply perfect population detection.

Method	TPR (%) $\uparrow$	Self-BLEU $\downarrow$	Repeated 4-grams (%) $\downarrow$	Distinct-3 $\uparrow$
Unwatermarked	—	0.01928 ^{0.02229} _{0.01649}	2.92 ^{4.33} _{1.96}	0.9514 ^{0.9620} _{0.9379}
PRC (ours)	97 ¹⁰⁰ ₉₃	0.01890 ^{0.02227} _{0.01581}	2.38 ^{3.16} _{1.68}	0.9561 ^{0.9651} _{0.9462}
Repeat fallback on				
SynthID depth 2	100 ¹⁰⁰ ₁₀₀	0.01987 ^{0.02418} _{0.01592}	3.28 ^{4.39} _{2.32}	0.9462 ^{0.9587} _{0.9326}
SynthID depth 10	100 ¹⁰⁰ ₁₀₀	0.02206 ^{0.02518} _{0.01909}	2.42 ^{3.14} _{1.83}	0.9554 ^{0.9635} _{0.9463}
SynthID depth 30	100 ¹⁰⁰ ₁₀₀	0.02890 ^{0.03392} _{0.02412}	2.30 ^{3.09} _{1.58}	0.9589 ^{0.9683} _{0.9489}
TextSeal $\alpha = .1$	100 ¹⁰⁰ ₁₀₀	0.04727 ^{0.05421} _{0.04082}	2.74 ^{3.35} _{2.14}	0.9464 ^{0.9551} _{0.9380}
Gumbel-Max	100 ¹⁰⁰ ₁₀₀	0.20718 ^{0.23097} _{0.18343}	2.81 ^{3.71} _{2.05}	0.9456 ^{0.9557} _{0.9341}
Repeat fallback off				
SynthID depth 10	100 ¹⁰⁰ ₁₀₀	0.02138 ^{0.02534} _{0.01775}	9.35 ^{12.91} _{6.07}	0.8880 ^{0.9211} _{0.8528}
TextSeal $\alpha = .1$	100 ¹⁰⁰ ₁₀₀	0.03770 ^{0.04440} _{0.03157}	32.17 ^{36.95} _{27.15}	0.6603 ^{0.7092} _{0.6139}
Gumbel-Max	100 ¹⁰⁰ ₁₀₀	1.00000 ^{1.00000} _{1.00000}	51.73 ^{59.43} _{44.12}	0.4720 ^{0.5460} _{0.3978}